

CV

Basics

Name Luke Lowery
Email lukel@tamu.edu
Website <https://lukelowry.github.io>
Summary Currently employed at Texas A&M University. EE PhD Student of Power Systems

Work

2024 - Present	Research Assistant Texas A&M University • Develops Phasor & EMT models to accelerate the integration of renewables in modern transmission systems. • Performs novel research on the impact of Geomagnetic Disturbances on the nations electric grid. • Applies modern mathematics to solve open problems in energy and power research.
2023.05 - 2023.08	High Voltage Intern SpaceX • Switchgear & transformer installation, design, and coordination for mission-critical loads (5kV & 35kV) • Responsible for differential and over-current protection on critical feeder lines to production and launch-pad. • Designed & implemented 500 kW feeder and building distribution to a new production building in 4 weeks. • Co-designed the microgrid controller functionality with Tesla Engineers. Functionality allowed our Mega-Packs (smart battery storage) to create seamless transitions from the Utility to isolated island operation.
2022.08 - 2023.01	Power Systems Cooperative Education SpaceX • Coordinated with other SpaceX production and launch sites to create a standardized SCADA system. Performed transient analysis of synchronization anomalies to discover coordination errors. • Introduced solar panel expansion to decrease diesel use, turning Starbase into a 50% green energy site. • Designed a site-wide SCADA system to monitor and automate the site microgrid using Ignition as the front end.

2022.05 - 2022.08

SCADA Intern

SpaceX

- Automated various power distribution tasks such as load-shedding, interactive fuel-consumption projector, and predictive site-load calculator that is still utilized during launch operations.
- Deployed numerous PLCs, control panels, and sensors to increase grid visibility and control using Allen Bradley, Siemens, and SEL hardware. Communication protocols used: MQTT, DNP, Modbus, Profinet, GOOSE.

2021.05 - 2021.08

Control Systems Intern II

ICS Texas

- Designed and implemented control systems for numerous applications including natural gas sites, boiler systems, and production facilities. Primarily completed with Allen Bradley PLC hardware.

2020.05 - 2020.08

Control Systems Intern I

ICS Texas

- Mastered AutoCAD for the design of control panels that were manufactured in-house. Developed an automated AutoCAD program in LISP to synthesize to accelerate manufacturing.

Education

2023 - 2027

📍 College Station, TX

PhD

Texas A&M University

Electrical Engineering

2019 - 2023

📍 College Station, TX

Bachelor of Science

Texas A&M University

Electrical Engineering

Leadership

2025.05 - Present

Senator

Graduate and Professional Student Government

Student Senator for the ECEN Graduate Department

- Draft legislation that upholds university values and improves the Aggie experience.
- Vote and represent the Department of Electrical Engineering on University policies and procedures.

Publications

- 2026 **System Frequency Nadir and Trajectory Prediction With Discontinuity Constraints in Governor Dynamics**
Jongoh Baek, *Luke Lowery*, and Adam B. Birchfield
IEEE Open Access Journal of Power and Energy, 13, pp. 274–285
DOI: 10.1109/OAJPE.2026.3675094
- 2026 **Spectral Graph Filters for Large Power Systems**
Luke Lowery and Adam B. Birchfield
2026 IEEE Texas Power and Energy Conference (TPEC), pp. 1–6
DOI: 10.1109/TPEC67884.2026.11513110
- 2026 **Using Spectral Graph Wavelets to Analyze Large Power System Oscillation Modes**
Luke Lowery, Jongoh Baek, and Adam Birchfield
59th Hawaii International Conference on System Sciences (HICSS)
- 2025 **EMT Simulation with Spectral Graph Wavelets**
Luke Lowery and Adam B Birchfield
2025 IEEE Texas Power and Energy Conference (TPEC), pp. 1–6
- 2025 **Voltage Stability in Plausible Non-Uniform Geomagnetic Disturbances**
Luke Lowery and Adam B. Birchfield
IEEE Transactions on Power Systems, pp. 1-12
DOI: 10.1109/TPWRS.2025.3588826
- 2024 **Voltage Stability Interface Limits in Non-Uniform Geomagnetic Disturbances**
Luke Lowery and Adam B Birchfield
2024 56th North American Power Symposium (NAPS), pp. 1–6

Presentations

- 2024.03.01 **Graduate Speaker**
 College Station, Texas ECEN Townhall
- 2024.03.01 **Graduate Speaker**
 College Station, Texas ECEN Fellowship Banquet
- 2024.03.01 **Speaker**
 College Station, Texas Modern Methods in Electromagnetic Transient Simulations
- 2024.03.01 **Speaker**
 College Station, Texas Voltage Stability in Non-Uniform Geomagnetic Disturbances

Awards

2025.12.01	Best Paper 59th Hawaii International Conference on System Science <i>Best Paper in Electric Energy Systems Track</i>
2025.11.01	Department Poster Award Texas A&M University <i>2nd Place in Department Poster Event</i>
2024.03.01	Michael Powell Graduate Fellowship Texas A&M University
2024.03.01	Thomas Powell 1962 Fellowship Texas A&M University

Projects

2025.12 - 2026.02	Blackout USA <i>An educational tool for all backgrounds to understand the challenges of operating the electric grid.</i>
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